GG-C-611C
September 12, 1975
SUPERSEDING
GG-C-00611B (DSA-DM)
October 19, 1973

FEDERAL SPECIFICATION

# COUNTING APPARATUS, BACTERIAL COLONY AND PLATE, COUNTING, BACTERIOLOGICAL

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for use of all Federal agencies.

## 1. SCOPE

1.1 This specification covers a Quebec type counting apparatus, including a Wolffhuegel counting plate, suitable for counting bacterial colonies.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

Federal Specifications:

| | |
|---|---|
| PPP-B-566 | Boxes, Folding, Paperboard. |
| PPP-B-585 | Boxes, Wood, Wirebound. |
| PPP-B-601 | Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | Boxes, Shipping, Fiberboard. |
| PPP-B-676 | Boxes, Setup. |

Federal Standards:

| | |
|---|---|
| FED-STD-123 | Marking for Shipment (Civil Agencies). |

(Activities outside the Federal Government may obtain copies of Federal specifications and standards as outlined under General Information in the Index of Federal Specifications and Standards, at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal specifications and standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

| | |
|---|---|
| MIL-P-116 | Preservation, Methods of. |
| MIL-M-7298 | Manual, Technical: Commercial Equipment. |
| MIL-L-10547 | Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible. |

Military Standards:

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | Marking for Shipment and Storage. |

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

American Society for Testing and Materials (ASTM) Standard:

| | |
|---|---|
| ASTM C162 | Standard Definitions of Terms Relating to Glass and Glass Products. |

(Copies may be obtained upon application to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)



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3. REQUIREMENTS

3.1 Material. The material shall be suitable in every respect for the purpose intended and shall be as herein specified.

3.2 Design, construction and dimensions. The counting apparatus shall consist of the housing, a lens, a lamp, an electric cord, and a counting plate. Figure 1 shows a typical design for the counting apparatus, and Figure 2 shows the design of the bacteriological counting plate (see 6.2).

3.2.1 Housing. The housing shall be fabricated from sheet metal. The housing shall be 10 ± 1/4 inches (in.) long, 10-3/4 ± 1/4 in. wide, and 10-1/4 ± 1/4 in. high. The top surface of the housing shall be inclined at an angle, and shall contain an opening 4 ± 1/8 in. in diameter through which the counting plate shall be seen.

3.2.2 Lens. The counter shall be equipped with a lens 4-1/2 ± 1/8 in. in diameter, having a focal length of 12.00 ± 0.60 in. and a magnification of 1.5X. The lens shall be mounted on a sliding rod for focusing by the operator, and shall be constructed so that it may be pushed out of the way when not in use. The focusing range shall be from 1-1/4 in. to 5-1/4 in. from the object. The focal length characteristics shall be tested as specified in 4.4.1.

3.2.3 Lamp and cord. The counter shall be equipped with a 40-watt lamp and an electric connecting cord. The lamp shall be readily accessible for replacement.

3.2.4 Counting plate. One Wolffhuegel counting plate (see 6.2) shall be furnished with each colony counter. The plate shall be glass, 3.2 ± 0.1 millimeters (mm.) thick, and shall be square with sides 114 mm. in length (angular tolerance ± 2°). The plate shall have a lined area as shown in Figure 2. The area shall be square with 100-mm. sides (angular tolerance ± 1°) and shall contain 100 squares with 10-mm. sides (angular tolerance ± 1°). Each of the diagonal squares shall be subdivided into 9 equal squares (angular tolerance ± 1°). The lines shall be engraved or etched or silkscreened. Lines which are engraved or etched shall be 0.25 ± 0.05 mm. wide. Lines which are silkscreened shall be 0.25 + 0.15, -0.10 mm. wide. Area tolerances of the small squares shall be ± 5%. The center of the lined area shall be not more than 1.0 mm. from the center of the plate, and the sides of the lined area shall be parallel with the sides of the plate (angular tolerance ± 2°).

3.3 Centering device. A centering device, such as centering screws, shall be provided on the case surface to facilitate the centering of Petri dishes when circularly ruled plates are used.



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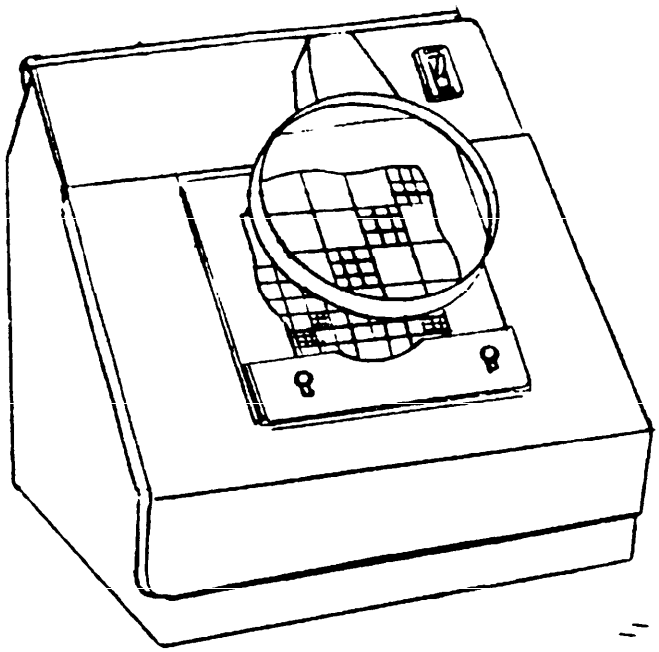


FIG. 1 COUNTING APPARATUS BACTERIAL COLONY (TYPICAL)

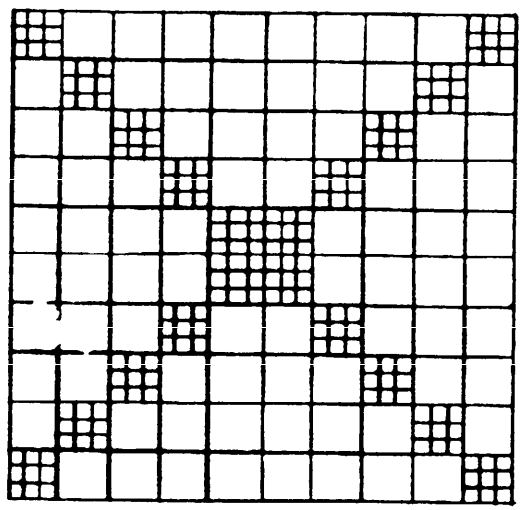


FIG. 2 PLATE, COUNTING, BACTERIOLOGICAL



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3.4 Tilting device. An auxiliary tilting device shall be provided. The device shall be suitable for providing change of front panel slope as required by the operator.

3.5 Finish. Unless otherwise specified, the manufacturer's standard finish shall be furnished.

3.6 Identification marking. Each colony counting apparatus shall be permanently and legibly marked with the name or trademark of the manufacturer.

3.7 Service data. The contractor shall furnish with each unit two copies each of a technical manual which contains complete instructions for installation, operation and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for identical item within two years preceding those previously accepted, the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the contracting officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

3.8 Workmanship. Workmanship shall be first class throughout. The finished colony counting apparatus shall be free from defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I.

TABLE I. Sampling for examination

| Unit of product | Type of defect | Inspection level | AQL (percent defective) |
|---|---|---|---|
| | Visual examination | | |
| Counting apparatus | Major A defects | II | 1.0 |
| Glass components | Major B defects | II | 1.0 |
| Counting apparatus | Minor A defects | II | 2.5 |
| Glass components | Minor B defects | II | 2.5 |
| Counting apparatus | Dimensional examination | S-2 | 2.5 |

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II.

TABLE II. Sampling for tests

| Component (unit of product) | Characteristic | Requirement | Test procedure | Inspection level | AQL |
|---|---|---|---|---|---|
| Counting apparatus | Focal length | 3.2.2 | 4.4.1 | S-1 | * |

* Acceptance number shall be zero.

4.3 Examination. The counting apparatus shall be examined to determine compliance with all requirements contained in this specification.



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4.3.1 Classification of defects. Examination shall be conducted in accordance with the following table:

TABLE III. Classification of defects

| Categories | Defects |
|---|---|
| Major A | |
| 101 | Component part missing. |
| 102 | Counting apparatus not free of sharp edges. |
| 103 | Painted surfaces not free of blisters. |
| 104 | Plated surfaces not completely plated. |
| 105 | Plating of plated surfaces does not adhere. |
| 106 | Marking on counting plate not complete, correct, legible or permanent. |
| Major B | |
| 151 | Component not free of checks. |
| 152 | Component not free of chips. |
| 153 | Component not free of splits. |
| 154 | Component not free of blisters. |
| 155 | Component not free of blowover. |
| 156 | Component not free of protrusions. |
| Minor A | |
| 201 | Metal surfaces not free of pits or burs. |
| 202 | Manufacturer's identification marking not present, complete, correct, legible or permanent. |
| Minor B | |
| 251 | Component not free of stones. |
| 252 | Component not free of seeds. |
| 253 | Component not free of tears. |
| 254 | Component not free of bloom. |
| 255 | Lens not free of striae. |
| 256 | Lens not free of sleek. |

Examination shall not be restricted to the classified possible defects listed above.

Glass defects are as defined in ASTM Designation C162.



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4.3.2 Dimensional examination. The counting apparatus shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Focal length. The lens of the counting apparatus shall be tested to determine focal length. An optical bench accurate to within 5% shall be used.

4.5 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The Inspection Level shall be S-2, and the Acceptable Quality Level shall be 4.0 percent defective.

TABLE IV. Examination of preparation for delivery

| Examination | Defect |
|---|---|
| Containers | Not as specified. |
| Quantity | Quantity in interior packages and shipping containers not as specified. |
| Materials | Any component missing or damaged. |
| Workmanship | Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted. |
| Markings | Omitted, incorrect, illegible, improper size, location, sequence, or method of application. |

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.1).

5.1.1 Unit of issue. One counting apparatus or one plate, as specified, constitutes one unit of issue.



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5.1.2 Level A.

5.1.2.1 Unit package.

5.1.2.1.1 Counting apparatus with plate. Each unit shall be thoroughly cleaned and packaged in accordance with MIL-P-116, method IC-2.

5.1.2.1.2 Plate only. Each unit shall be wrapped in tissue or banier material and packaged in a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable, or PPP-B-636, class domestic. Closure shall be adequate to prevent accidental opening.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. One counting apparatus or 24 plates shall be packed in box designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with the appendix of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Exterior container. One counting apparatus or 24 plates shall be packed in a box designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1, or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.



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5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with FED-STD-123. The unit container shall be the container specified in 5.1.2.1 or 5.1.3, and the shipping container shall be the container specified in 5.2.1.1, 5.2.2.1 or 5.2.3.

5.3.2 Military activities.

5.3.2.1 Interior packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number and date of this specification.
b. Counting apparatus complete (see 3.2).
c. Wolffhuegel counting plate (see 3.2.4).
d. Other than standard finish (see 3.5).
e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following items as listed in the Federal Supply Catalog:

| National Stock No. | Item Identification |
|---|---|
| 6640-00-418-8200 | COUNTING APPARATUS, BACTERIAL COLONY, Quebec, 115 volt, AC-DC |
| 6640-00-418-8205 | PLATE, COUNTING, BACTERIOLOGICAL, Wolffhuegel |



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Navy - MS
Air Force - 03

Civil Agency Coordinating Activities:

HEW - HSM, NIH
VA - DMS

Preparing activity:

DSA - DM

Project No. 6640-0910

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CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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